SUPSI
University of Applied Sciences and Arts of Southern Switzerland

Designing With

A New Educational Module to Integrate Artificial Intelligence, Machine Learning and Data Visualization in Design Curricula

+ Didactic Guidelines

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Contents

3.1 Sub-module 1: Getting familiar with Machine Learning (ML)

This sub-module teaches students to train simplified ML models to create design artefacts {ML-Obj}. Students develop procedural knowledge through practical guided activities (e.g. RunwayML step-by-step tool tutorials) and they are asked to apply the acquired procedure to a familiar task (e.g. creating a logo). Students are first provided with a basic literacy of ML including technical vocabulary {ML-Act1} and primary functionalities {ML-Act2}. This activity is meant to promote factual and conceptual knowledge by teaching the basic elements of the discipline and their interrelationship to explain the functionality of ML tools. Once basic literacy is introduced, specific case studies are provided to contextualise the application of ML within the design practice {ML-Act3}. At this point, the teaching moves from a theoretical to a procedural level. In this direction, students are first introduced to a guided tutorial {ML-Act4.3} on how to use an ML tool (e.g. RunwayML) starting from setting a design goal {ML-Act4.1} and creating their dataset {ML-Act4.2}. Then, to familiarise themselves with the procedure, students individually apply what they learned by training an ML model aligned with the design goal {ML-Act5}. Lastly, moving the focus to meta-cognitive knowledge, students are asked to analyse and document the process by breaking down the steps and selecting the relevant results {ML-Act6}. This activity aims to create awareness and knowledge of cognition. To foster participation and sharing of results, students are asked to present their work and question others {ML-Act7} (Fig. 7).

Educational Objectives

{ML-Obj}
Students will learn to | train a Machine Learning model | for a specific design purpose

Instructional Activities

{ML-Act.1}
Activity intended to | provide students | with vocabulary knowledge of ML

{ML-Act.2}
Activity intended to | provide students | with basic functionality of ML

{ML-Act.3}
Activity intended to | provide students | with practical examples of ML applied to design

{ML-Act.4}
Activity intended to | explain | training a simplified ML model

{ML-Act.4.1}
Activity intended to | set or align | with a design goal

{ML-Act.4.2}
Activity intended to | introduce students | to familiarising, collecting or creating data

{ML-Act.4.3}
Activity intended to | provide students | a step-by-step tool tutorial, such as RunwayML

{ML-Act.5}
Activity intended to | allow students | to individually train a simplified MLmodel for a design goal

{ML-Act.6}
Activity intended to | provide students | with a structured template for process and results documentation

{ML-Act.7}
Activity intended to | make students | explaining and questioning the process

Sub-module 1: Getting familiar with ML	Cognitive process dimension					
	(1) Remember	(2) Understand	(3) Apply	(4) Analyze	(5) Evaluate	(6) Create
(a) Factual Knowledge		{ML-Act.1}				
(b) Conceptual Knowledge		{ML-Act.2} {ML-Act.3}				
(c) Procedural Knowledge		{ML-Act.4} {ML-Act.4.1, 4.2, 4.3}	{ML-Act.5} {ML-Obj}			
(d) Metacognitive Knowledge				{ML-Act.6}	{ML-Act.7}	

Fig. 7 Sub-Module 1: 'Getting Familiar' with Machine Learning. Educational Objectives and Instructional Activities arranged into the Bloom Taxonomy Table.

6. References

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